

Low-Power RISC-V and FPGA-Based SoC Architectures for IoT Devices: A Literature Review

Abstract.

The rapid growth of the Internet of Things (IoT) and edge computing has created a strong demand for devices that are not only **energy-efficient** but also **computationally capable** of handling increasingly complex tasks under strict power constraints. Designers are now challenged to meet the requirements of advanced, real-time applications while keeping energy budgets extremely low — a crucial factor for battery-powered and energy-harvesting systems. This paper aims to consolidate recent research and provide a **unified perspective** on how open-source hardware, power optimization, and on-device intelligence are shaping the next generation of IoT systems.

This paper presents a **literature review** of key technological approaches that address these challenges across different layers of the design stack, including **hardware architecture, power management, and embedded intelligence**. First, we examine **X-HEEP**, an open-source and highly configurable **RISC-V microcontroller platform** that enables flexible integration of custom hardware accelerators while maintaining ultra-low-power operation. Second, we review the concept of **Dynamic Voltage and Frequency Scaling (DVFS)** and its architectural simulation models, which help designers predict power–performance trade-offs and optimize energy efficiency during the early stages of system development. Third, we analyze the emerging field of **Tiny Machine Learning (TinyML)**, which enables advanced machine learning models to run directly on constrained devices, supporting intelligent sensing and real-time decision-making without relying on continuous cloud connectivity.

By combining these perspectives, this review highlights the **convergence of open architectures, predictive power modeling, and lightweight machine learning** as a pathway toward next-generation IoT systems. While each approach tackles specific design challenges — configurability, early power optimization, and edge intelligence — integrating them remains difficult due to **tool fragmentation, lack of standard benchmarks, and heterogeneous hardware ecosystems**. The review also outlines current **research gaps**, including the need for **hardware–software co-design, energy-aware AI benchmarking, and unified development frameworks** that simplify the creation of low-power, intelligent edge devices.

Beyond academic relevance, these trends have a profound impact on **industry and innovation**, enabling the development of smart cities, autonomous systems, and sustainable connected solutions. By

synthesizing insights from recent advances in **open-source hardware**, **power optimization strategies**, and **edge AI frameworks**, this paper provides a **clear roadmap for both researchers and engineers**.

It aims to guide future developments in designing **flexible, energy-efficient, and intelligent embedded systems** capable of meeting the performance and power demands of the rapidly evolving IoT landscape.

Introduction.

The **Internet of Things (IoT)** has rapidly transformed from a futuristic concept into a practical reality, with billions of interconnected devices now collecting and processing data in homes, cities, factories, and healthcare environments. These devices operate at the very edge of the network, often under strict **power and cost constraints**. Many IoT nodes run on small batteries or harvest energy from their surroundings, making **energy efficiency a critical design requirement**. At the same time, modern applications — such as predictive maintenance, environmental monitoring, smart agriculture, and wearable health devices — increasingly require local computation and even **machine learning (ML) capabilities** to deliver real-time responses, preserve privacy, and reduce network bandwidth.

Meeting these conflicting demands — **high computational capability versus ultra-low power operation** — is one of the most important challenges for embedded system designers today. Traditional closed, fixed-function microcontrollers can no longer keep up with the rapidly evolving requirements of IoT applications. This has led to the rise of **open hardware architectures**, such as RISC-V, which enable more flexible and customizable solutions. Developers can now extend these architectures with **hardware accelerators** or specialized processing blocks tailored to specific tasks, improving both performance and energy efficiency.

At the same time, **power management techniques** are becoming increasingly sophisticated. Approaches like **Dynamic Voltage and Frequency Scaling (DVFS)** allow systems to dynamically adapt their energy consumption based on workload, helping designers reach optimal trade-offs between performance and battery life. Accurate early-stage

modeling of these techniques enables better design decisions before committing to expensive hardware manufacturing.

Another important development is the rise of **Tiny Machine Learning (TinyML)** — a paradigm that brings intelligence directly to resource-constrained devices. By running compact and optimized ML models on low-power microcontrollers, TinyML reduces latency, improves privacy, and minimizes dependence on cloud infrastructure. This trend is reshaping the IoT landscape by enabling smarter, more autonomous devices that can make decisions locally.

This paper presents a **literature review** of three key technological directions shaping the future of ultra-low-power IoT systems: **open and configurable RISC-V architectures, power-aware design methodologies such as DVFS, and TinyML for on-device intelligence**. By analyzing these works together, we aim to provide a clear picture of how hardware flexibility, energy-aware design, and lightweight machine learning are converging to build the next generation of **smart and energy-efficient edge devices**.

RISC-V and Low-Power Architectures.

Edge computing has experienced rapid growth as more Internet of Things (IoT) devices demand real-time data processing with very limited energy budgets. Despite this expansion, most current edge platforms struggle to achieve both high performance and ultra-low power operation.

Conventional microcontrollers are often fixed in design and cannot be easily adapted to new workloads or specialized hardware accelerators without major hardware redesign. Recent research has therefore focused on heterogeneous System-on-Chip (SoC) architectures that combine general-purpose processors with domain-specific accelerators, aiming to improve computing efficiency while keeping energy consumption minimal. However, many of these solutions remain hard to customize because they require complex and time-consuming modifications to register-transfer level (RTL) hardware.

The **X-HEEP** platform addresses these limitations by offering an open-source, highly configurable, and extendible RISC-V-based microcontroller tailored for ultra-low-power edge applications. It enables developers to explore different processor cores, interconnect bus

structures, and memory hierarchies while supporting fine-grained memory bank control to match the needs of custom accelerators. Furthermore, X-HEEP integrates advanced power management techniques such as clock gating and power gating, along with dedicated interfaces to control the power state of connected accelerators. A practical implementation, known as **HEEPocrates**, shows how the platform can be adapted for healthcare applications by incorporating coarse-grained reconfigurable array (CGRA) and in-memory computing (IMC) accelerators.

Silicon prototypes built using TSMC 65 nm low-power CMOS technology, as well as FPGA-based evaluations on Xilinx platforms, demonstrate that X-HEEP can operate across a wide voltage range (0.8–1.2 V) and achieve frequencies from 32 kHz up to 470 MHz. Its measured power consumption varies from just 270 μ W at the lowest operating point to 48 mW at maximum speed. Experimental results show competitive efficiency compared to other state-of-the-art microcontrollers and confirm that combining CGRA and IMC accelerators can yield up to **4.9× and 4.8× energy savings** compared to CPU-only execution. These results highlight X-HEEP’s potential as a flexible, energy-aware platform for future IoT and edge AI systems.

Dynamic Voltage and Frequency Scaling (DVFS).

Power management has become a central challenge in the design of modern embedded systems, which are increasingly complex and required to deliver high performance while operating under strict energy and cost constraints. This work explores low-power design strategies at the architectural level, with a particular focus on **Dynamic Voltage and Frequency Scaling (DVFS)** as one of the most effective techniques for reducing energy consumption while preserving performance. While register-transfer level (RTL) models offer high accuracy, they are costly and time-consuming during the early stages of system development. Architectural-level modeling and virtual prototyping, by contrast, provide a faster and more cost-efficient way to evaluate incomplete systems and guide critical design decisions before hardware implementation.

The paper reviews the role of architectural simulation in enabling early power–performance trade-off analysis, showing how energy management strategies—especially DVFS—can be integrated into simulation

platforms to improve design exploration. It highlights both hardware-based and software-based DVFS control models, noting that hardware-driven approaches typically offer faster response to performance requirements and lower latency when transitioning between voltage and frequency levels.

Additionally, the study discusses the application of DVFS in multi-core and real-time embedded systems, where task deadlines and priority constraints present significant scheduling challenges. It examines issues such as transition latency and its impact on meeting timing requirements, while presenting design considerations for achieving predictable performance. A case study is also presented to demonstrate how integrating hardware-assisted DVFS into early-stage simulation environments can provide practical insight for designing energy-efficient, high-performance embedded systems long before silicon fabrication.

Tiny Machine Learning (TinyML).

The rapid evolution of the Internet of Things (IoT) has created an urgent need for intelligent processing directly at the network edge, where devices are extremely resource-constrained and often operate on small batteries or energy-harvesting systems. Traditional cloud-based machine learning introduces latency, privacy concerns, and significant communication overhead, making it unsuitable for many real-time applications. In response, **Tiny Machine Learning (TinyML)** has emerged as a transformative paradigm that enables advanced machine learning models to run directly on ultra-low-power microcontrollers and embedded platforms. This approach reduces reliance on cloud connectivity, lowers energy consumption, and allows data to be processed closer to where it is generated.

This paper presents a systematic review of the TinyML ecosystem, analyzing its key technological components, recent progress, and open challenges. It explores the hardware platforms that support TinyML, ranging from simple ARM Cortex-M microcontrollers to specialized AI accelerators designed for milliwatt-level energy budgets. It examines software toolchains and frameworks such as **TensorFlow Lite Micro (TFLM)**, microTVM, and Edge Impulse, which simplify model deployment and optimize inference performance for constrained devices. The review also investigates training and quantization techniques that

reduce model size and computational complexity, making deep learning feasible within limited memory and processing power.

Beyond the core technology stack, this study highlights the growing role of **connectivity** and network architectures, including low-power wide-area networks (LPWAN), 5G, and hybrid cloud-edge models that enable scalable and secure IoT solutions. Several real-world applications are discussed, from industrial condition monitoring and predictive maintenance to smart agriculture, wearable health devices, and autonomous systems. Each use case demonstrates how TinyML enables on-device intelligence while maintaining extremely low energy consumption and cost efficiency.

Finally, the paper identifies key research gaps and future directions, such as the need for standardized benchmarking tools, better datasets for embedded AI, and improved interoperability across heterogeneous hardware. It calls for continued advances in compiler optimization, neural architecture search (NAS) for constrained devices, and security mechanisms that protect models and data at the edge. By synthesizing current knowledge and trends, this review provides a clear roadmap for researchers and practitioners aiming to design next-generation IoT solutions that combine **machine intelligence with ultra-low power operation** at scale.

Discussion.

The three studies reviewed in this paper reflect the **multi-layered complexity of designing ultra-low-power and intelligent embedded systems** for the Internet of Things (IoT). Each work approaches the problem from a different angle: **hardware architecture flexibility**, **power management and design-time modeling**, and **software-level intelligence**. Together, they paint a broad yet connected picture of how future IoT platforms can be both energy-efficient and computationally capable.

The first work, **X-HEEP**, proposes an **open-source and highly configurable RISC-V microcontroller platform** that allows designers to integrate custom hardware accelerators while maintaining ultra-low power operation. This is significant because closed, fixed-architecture microcontrollers often limit innovation and prevent tailored solutions.

With X-HEEP, researchers can experiment with different hardware blocks and reconfigure the system for diverse applications such as sensor fusion, signal processing, or lightweight machine learning. However, this flexibility also brings complexity: designers must carefully evaluate trade-offs in power, area, and timing to avoid inefficiencies when adding accelerators.

In contrast, the second study emphasizes **Dynamic Voltage and Frequency Scaling (DVFS)** and **early-stage architectural modeling** rather than proposing new hardware. This approach shows that saving energy is not just about hardware selection; it also depends on **intelligent power control strategies** applied from the earliest design phases. By simulating DVFS policies before chip fabrication, engineers can better understand how workloads will affect energy consumption and system responsiveness. This is especially important for **multi-core and real-time embedded systems**, where timing constraints must be balanced against aggressive power reduction. Still, the methodology relies heavily on accurate models and can add complexity to the design workflow — a barrier for small teams or rapid prototyping.

The third paper explores a different but increasingly important perspective: **Tiny Machine Learning (TinyML)**, which brings **artificial intelligence directly to the edge**. Instead of sending data to the cloud for processing, TinyML enables small, battery-powered devices to perform local inference using highly optimized neural networks. This reduces latency, improves privacy, and saves bandwidth, which is critical in remote or bandwidth-limited settings. The review shows how frameworks such as **TensorFlow Lite Micro (TFLM)** and emerging AI accelerators are enabling this shift. At the same time, it highlights real challenges: fragmented tools, limited standardized benchmarks, and hardware heterogeneity make deployment and comparison difficult. TinyML is still evolving, and the ecosystem lacks the mature co-design methodologies seen in other embedded system domains.

When these three directions are viewed together, a **clear convergence emerges**. Open and modular RISC-V platforms such as X-HEEP provide the **hardware foundation** for custom low-power designs. DVFS and power modeling offer **design-time energy optimization**, helping engineers make informed decisions before committing to silicon. TinyML then leverages this improved hardware to run increasingly capable AI models at the edge, enabling smarter and more autonomous devices. However, integrating these approaches is not yet seamless. Most TinyML frameworks do not fully exploit advanced power management like DVFS.

Power modeling tools are often unaware of the unique characteristics of machine learning workloads. And RISC-V SoCs, while flexible, still lack standardized interfaces for dynamic power control and accelerator management.

There is also a broader challenge of **tool fragmentation**. Developers must currently navigate separate ecosystems for RISC-V hardware design, DVFS modeling, and TinyML deployment. This slows down innovation and increases the barrier for smaller companies or academic teams. Another issue is the **lack of reliable benchmarking and evaluation tools** that can fairly compare power efficiency across different combinations of hardware and software. Without standard metrics, it is difficult to know which design choices truly improve energy efficiency for AI-driven IoT nodes.

Looking ahead, the next step for the field will likely be **tighter hardware–software co-design**. Future platforms could integrate built-in support for DVFS and other adaptive power strategies, while also offering seamless deployment pipelines for TinyML models. Improved simulation environments that include AI workloads and power management in one place would help researchers experiment more easily and predict real-world behavior with higher accuracy. Additionally, stronger **security and privacy measures** will be needed, as edge devices running AI become attractive targets for attacks.

Overall, the three studies collectively show that the future of ultra-low-power IoT lies in the **intersection of open architectures, smart energy management, and edge intelligence**. By closing the current gaps — especially in integration and tooling — designers will be able to create highly efficient and capable IoT systems that can last years on small batteries, process data locally for privacy and speed, and adapt dynamically to changing workloads.

Conclusion.

This review explored three different but highly complementary directions in the development of ultra-low-power and flexible embedded systems for the Internet of Things (IoT). The first work, **X-HEEP**, demonstrated how an open-source and configurable RISC-V microcontroller can serve as a versatile foundation for building experimental platforms and custom

accelerators while maintaining energy efficiency. The second study shifted the focus from hardware configurability to architectural-level power management, showing how **Dynamic Voltage and Frequency Scaling (DVFS)** and early-stage simulation can help designers predict and optimize power–performance trade-offs before committing to physical silicon. The third paper added a software and algorithmic dimension by presenting the rise of **Tiny Machine Learning (TinyML)**, which enables real-time intelligence directly on extremely constrained devices, reducing cloud dependency, latency, and communication costs.

Together, these works show how the future of embedded systems design is becoming increasingly integrated across hardware and software layers. Open RISC-V architectures make it possible to experiment with new hardware accelerators; early power modeling ensures that energy budgets can be managed effectively; and TinyML demonstrates that even the smallest edge devices can perform meaningful AI tasks while staying energy-efficient. When combined, these technologies have the potential to create a new generation of IoT systems that are smarter, more autonomous, and capable of operating for long periods on minimal power.

Looking forward, the convergence of open architectures, predictive energy modeling, and intelligent edge computing will be a key driver of innovation in IoT. Systems that can adapt their performance and power use in real time, while running advanced AI locally, will enable applications ranging from industrial automation and smart agriculture to healthcare monitoring and autonomous systems. For researchers and engineers, this means new opportunities to explore cross-layer optimization, combining flexible RISC-V hardware, robust energy management, and lightweight machine learning to push the limits of what ultra-low-power devices can achieve.

TinyML for Ultra-Low Power AI and Large-Scale IoT Deployments: A Systematic Review

- • **DVFS and Its Architectural Simulation Models for Improving Energy Efficiency of Complex Embedded Systems in Early Design Phase**

X-HEEP: An Open-Source, Configurable and Extendible RISC-V Microcontroller for the Exploration of Ultra-Low-Power Edge Accelerators